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Delayed Embryonic Development and Implantation in Senescent Golden Hamsters[1]

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Early embryonic stages from young (3–5 months) and senescent (14–15 months) golden hamsters were compared after 3, 3½, 4, 4½, 5, and 5½ days of pregnancy. Sections of epoxy resin-embedded uteri from 22 young (89 embryos) and 18 senescent (33 embryos) animals were examined using light microscopy. Morphologically, embryos from senescent hamsters exhibited a 12-h delay in the time of uterine attachment, orientation of the inner cell mass, formation of giant trophoblastic cells, reduction of uterine epithelium, and early embryonic development. The Pontamine Blue reaction in older uteri was delayed, sporadic, and less intense than in younger animals. Decidual cells first appeared in the uteri of senescent animals 12 h later, in accordance with the delay of the implanting blastocyst. The extent of the decidual cell reaction in the majority of senescent hamsters appeared similar to that in younger animals 12 h earlier.

Reproductive senescence in several mammals is characterized by a prolonged period of gestation. This has been reported for goats (Asdell, 1929), sheep (Terrill and Hazel, 1947), cows (Brakel *et al.*, 1952), and mice (Roman and Strong, 1962). Soderwall and co-workers (1960) found that golden hamsters, 14 months of age, exhibit a mean gestation period approximately 24 h longer than hamsters 3–5 months of age.

Most polytocous laboratory mammals exhibit a decline in litter size with an increase in age (Ingram *et al.*, 1958; Biggers *et al.*, 1962; Finn, 1962; Blaha, 1964; Adams, 1970). In the female golden hamster, a pronounced drop in fecundity occurs at 14 months of age (Soderwall *et al.*, 1960). The number of ova produced is not the immediate cause for such a decline, as the number of corpora lutea remains the same in young and senescent hamsters (Thorneycroft and Soderwall, 1969; Connors *et al.*, 1972). Although the viability of the ova may be a limiting factor (Henderson and Edwards, 1968), the uterus is generally considered to be the primary cause for reduction of litter size among older animals (Talbert and Krohn, 1966; Biggers, 1969; Finn, 1970).

A preliminary study (Connors, 1969) revealed an apparent developmental lag during the early stages of implanting senescent hamster blastocysts. The present study was undertaken to critically analyze the early morphological stages of implantation in young and senescent golden hamsters to evaluate possible causes for implantation failure.

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MATERIALS AND METHODS

Golden hamsters (*Mesocricetus auratus*), which were housed singly under conditions of constant temperature (21°–23°C) and lighting (12 h light, 12 h darkness), were maintained on a diet of Purina Laboratory Chow and lettuce greens daily. The majority of animals was taken from the colony maintained at the University of Oregon; however, some senescent females were retired breeders (10

TABLE 1
NUMBER OF GOLDEN HAMSTERS AND EMBRYOS AT EACH STAGE OF PREGNANCY AND THE PONTAMINE BLUE REACTION

Days post-ovulation	Young			Senescent		
	No. animals	Blastocysts examined	Pontamine Blue	No. animals	Blastocysts examined	Pontamine Blue
3	3	11	−	3	5	−
3½	4	12	+	3	3	−[a]
4	5	18	++	3	11	+[b]
4½	4	19	+++	3	5	+[b]
5	3	13	+++	3	4	++
5½	3	16	+++	3	5	++[c]
Total	22	89		18	33	

[a] One horn of one animal exhibited a very faint blue area.
[b] One animal exhibited a few strong blue areas (+++) as well as faint blue areas (+).
[c] One animal did not exhibit any blue areas, although a blastocyst was found during sectioning of the uterus.

months of age) purchased from Hilltop Lab Animals (Scottdale, Pennsylvania).

The vaginal discharge of young virgin females (3–5 months of age) and senescent multiparous females (14–15 months of age) was checked (Orsini, 1961) at least twice prior to mating to assure that they had normal cycling patterns. They were then placed with young males (3–6 months of age) of proven fertility in the early evening (1800) and observed for the onset of estrus by the lordosis response and copulation. Developmental time was dated from the time of ovulation, which was considered to be either 8 or 9 h from the onset of estrus (Harvey *et al.*, 1961).

Females pregnant for 3, 3½, 4, 4½, 5, and 5½ days (Table 1) were anesthetized and injected with 0.25 cm^3 of a 1% solution of Pontamine Blue in normal saline (Orsini, 1963) via the femoral vein. After 15 min the uteri were examined and the Pontamine Blue reaction recorded as: − (negative), + (faintly positive), ++ (moderately positive), and +++ (strongly positive). Fixation occurred by perfusion via the abdominal aorta with either cold 2.5% glutaraldehyde (phosphate buffered, pH 7.4) or 3.0% glutaraldehyde (cacodylate buffered, pH 7.2). Blue areas indicating implantation sites were dissected out and placed in the same glutaraldehyde fixative for an additional 1½–2 h. If blue areas were not evident, as normally occurs prior to implantation, the entire uterus was removed and sectioned into smaller pieces for further fixation. After the tissues were rinsed in buffer, they were sliced transversely with a razor blade into smaller pieces 3 mm long × 2 mm wide × 2 mm thick and postfixed in 2% osmium tetroxide with the appropriate buffer for 2 h. The tissues were subsequently dehydrated, embedded in Epon-Araldite epoxy resin (Mollenhauer, 1964), and sectioned at 1–3 μm on a Porter-Blum ultramicrotome. Serial sections were cut dry with glass knives, floated on water on glass slides, and dried on a slide warmer.

A total of 22 young and 18 senescent hamsters were utilized during the study (Table 1), with each time interval including animals dated 8 and 9 hours postestrus. Sections of 89 young blastocysts and 33 senescent blastocysts were examined at the early stages of pregnancy and photographed unstained using phase microscopy.

RESULTS

Implantation in the golden hamster has been described by Graves (1945), Ward (1948), and, most recently, by Butler (1970). In determining the age of the conceptus, Graves and Butler used postcoital age, and Ward used developmental age (based on ovulation at 0100 during the evening of copulation). By interpolating these authors' time periods of implantation and development of the young blastocyst, it has been found that their results are consistent with those in this study. The only difference occurs during the beginning stages of blastocyst attachment. In this study the uterine folds have tightly enclosed the blastocyst at an earlier time. This phenomenon is undoubtedly due to the superiority of glutaraldehyde as a fixative, which eliminates the shrinkage artifacts of these early stages that result from conventional fixatives.

The major difference occurring between young and senescent hamsters is the delayed development exhibited by blastocysts from older animals (Figs. 1–12). This is evidenced by the timing of the Pontamine Blue reaction, time of initial attachment of the blastocyst, orientation of the inner cell mass, first formation of giant trophoblast cells, appearance and development of decidual cells, reduction of the uterine epithelium, and early embryonic development.

The Pontamine Blue reaction becomes faintly visible in the young uterus at 3½ days of pregnancy (Table 1). This dye does not appear until the zona pellucida is lost and is associated with increased uterine capillary permeability (Orsini, 1963). The intensity of Pontamine Blue increased in the uterine horns of young animals at 4 days of pregnancy, and strongly positive blue areas were found from 4½ through 5½ days of pregnancy. In comparison, the uterus of senescent hamsters was inconsistent in response to the dye. Faint blue areas did not appear until day 4 of pregnancy and the same intensity was retained through 4½ days. One animal at each time interval exhibited a few strong blue areas in addition to the faint areas (Table 1). At days 5 and 5½ of pregnancy the Pontamine Blue reaction was similar to that of the young animal at 4 days, although blood lacunae surrounding the developing embryo at these stages were evident (Figs. 10 and 12). The total number of blue areas was considerably less in the older animals from 4½ to 5½ days of pregnancy (46 in senescent compared to 123 in young hamsters). However, one normal-appearing embryo was found in the uterine horn of a 5½-day senescent animal, which had not revealed any blue areas. Most of the uterine tissue not exhibiting blue areas was not sectioned, and implants from young and senescent animals may have gone undetected.

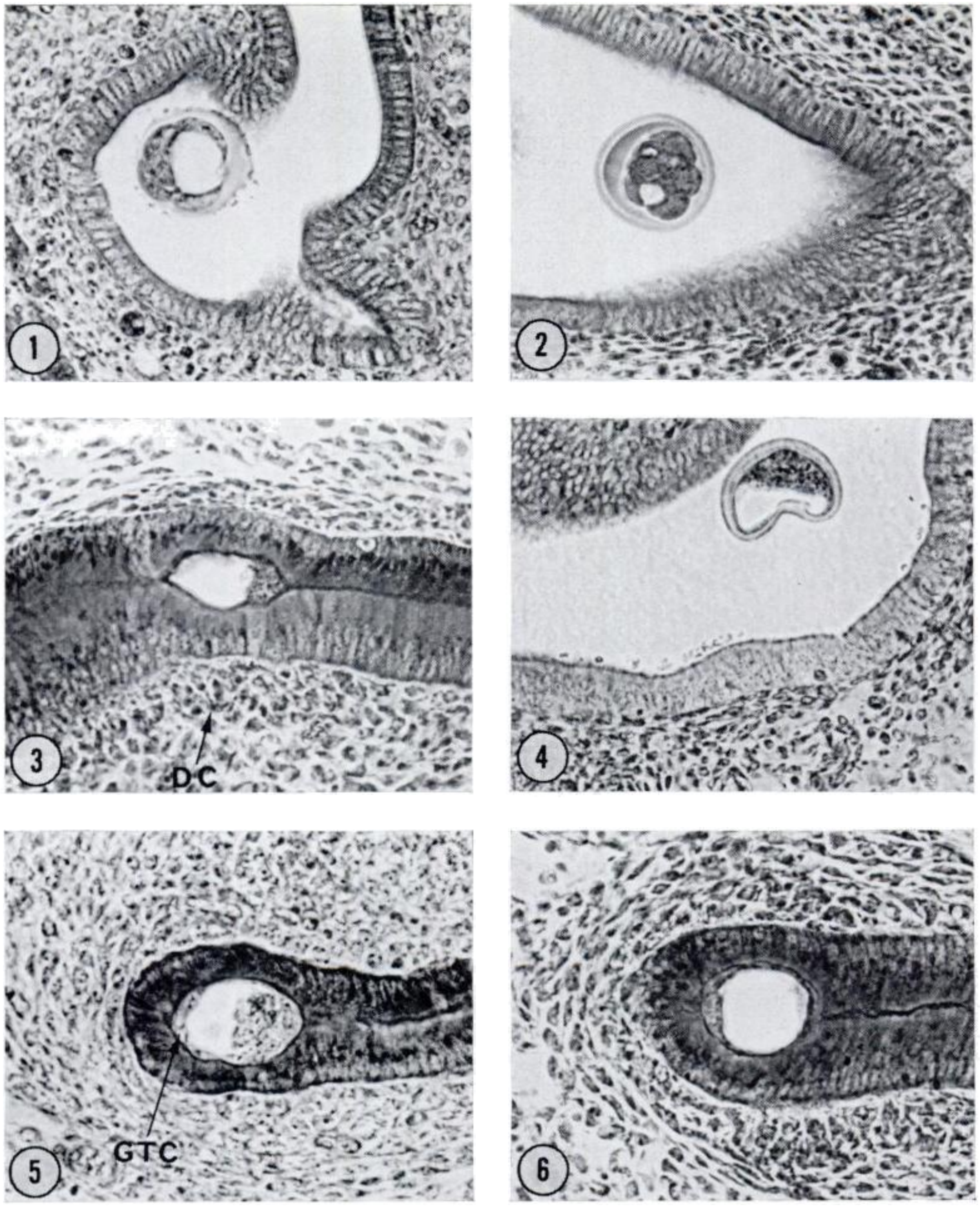
Some variation in development occurred among blastocysts from the same animal, especially in the younger animals. All but one blastocyst was enclosed by uterine epithelium at 3½ days of pregnancy, whereas all the blastocysts from the older animals were still free within the lumen (Figs. 3 and 4).

Orientation of the inner cell mass toward the antimesometrial position was not completed by many of the young blastocysts at 3½ days (6 of 12), but was correctly positioned in all blastocysts at 4½ days. In comparison, senescent blastocysts exhibited a delay. At 4 days the inner cell mass was not correctly orientated in most blastocysts (9 of 11), but was located antimesometrially in all uteri at 5 days of pregnancy.

The first appearance of giant trophoblastic cells at the abembryonic pole occurred on 3½ days of pregnancy in young blastocysts, whereas in senescent blastocysts they were not evident until day 4. In the mouse (Dickson, 1966) these cells appear shortly after the loss of the zona pellucida and apparently play an important role in implantation (Eaton and Green, 1963). The number of giant trophoblastic cells in the hamster appeared to be similar in young and old blastocysts.

The decidual cell response was also delayed in the senescent animal. While localized cells first appeared in the young animal at 3½ days (Fig. 3), they were absent in the senescent animal until day 4 (Fig. 6). The majority of senescent animals exhibited a decidual cell reaction similar to that noted in younger animals, which occurs 12 h earlier. However, the number of decidual cells extending mesometrially was fewer around three embryos at 5 and 5½ days of pregnancy when compared with the reaction surrounding young implants at 4½ and 5 days of pregnancy.

Reduction of the uterine epithelium followed the same trend in the two age groups. At 4½ days in the young animal the epithelium began to disappear (Fig. 7), but was unaffected until day 5 in the senescent hamster (Fig. 10). This delay



FIGS. 1–12. Transverse sections were photographed unstained using phase microscopy. In those photographs where the blastocyst is enclosed by uterine folds the mesometrial side is on the right.

FIG. 1. Three-day young hamster blastocyst prior to implantation. Note the prominent blastocoel. 300×.

FIG. 2. Three-day senescent hamster blastocyst prior to implantation. The blastocoel is considerably smaller than that in the 3-day young blastocyst. 300×.

FIG. 3. Three-and-a-half-day young blastocyst, which has just implanted. The inner cell mass is already correctly positioned on the mesometrial side. The decidual cells (DC) are just starting to develop on the sides of the uterine epithelium. 345×.

FIG. 4. Three-and-a-half-day senescent blastocyst still free within the uterine lumen. 300×.

FIG. 5. Four-day young blastocyst exhibiting giant trophoblastic cells (GTC) at the abembryonic pole. The decidual cell reaction completely surrounds the new implant. 345×.

FIG. 6. Four-day senescent blastocyst, in which the inner cell mass is still located antimesometrially. The decidual cell reaction is not nearly so extensive as that in the younger animal at the same stage of pregnancy. 345×.

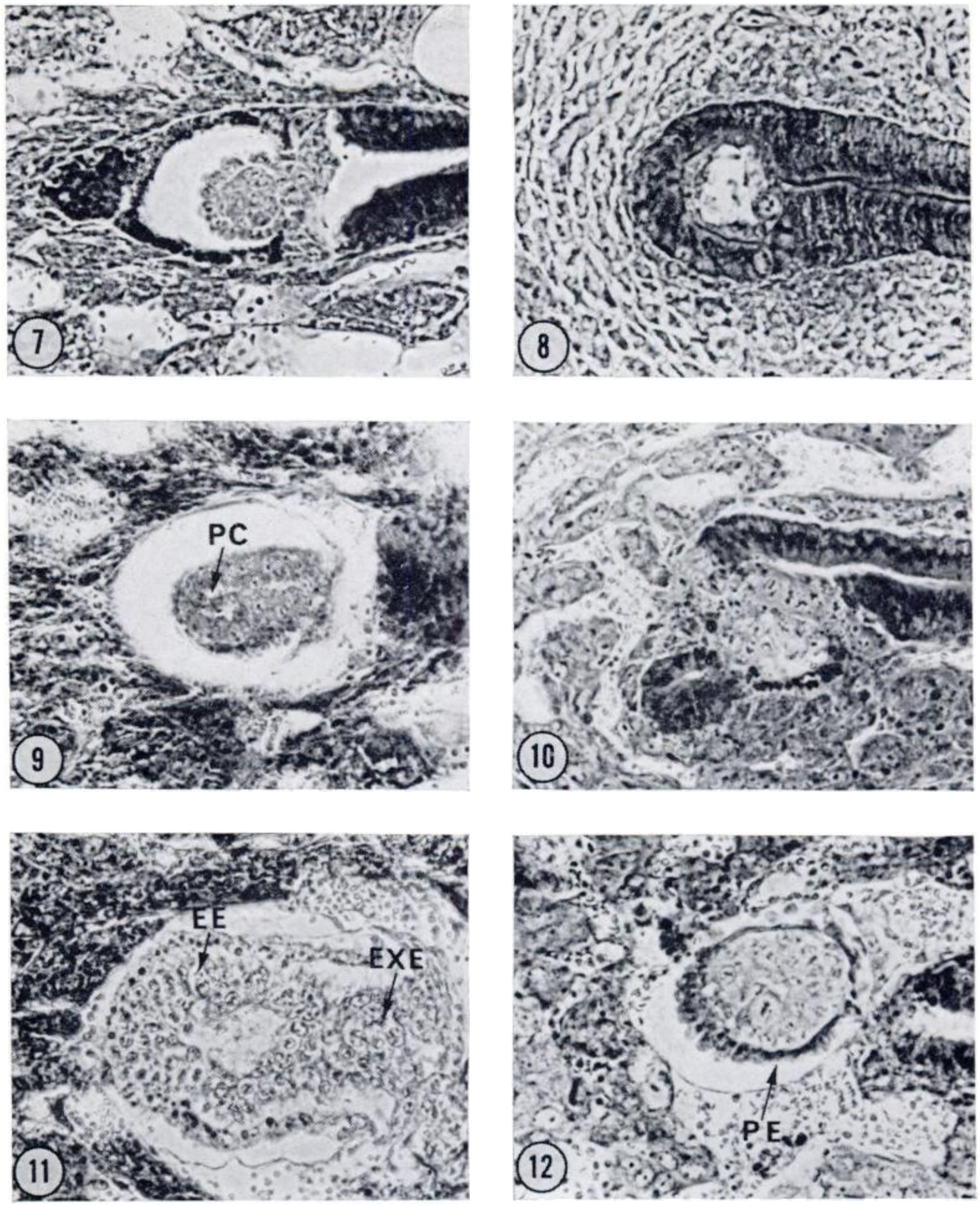


FIG. 7. Four-and-a-half-day young blastocyst in which the uterine epithelium is in a stage of reduction. 370×.

FIG. 8. Four-and-a-half-day senescent blastocyst. The inner cell mass and two giant trophoblastic cells are positioned mesometrially. The decidual cell reaction is comparable to that in the young animal at 4 days of pregnancy. 345×.

FIG. 9. Five-day embryo. The uterine epithelium has disappeared and the proamniotic cavity (PC) is starting to appear. 370×.

FIG. 10. Five-day senescent blastocyst. The uterine epithelium has started to disappear. Blood lacunae are now present such as that in the younger animal at 4½ days of pregnancy. 370×.

FIG. 11. Five-and-a-half-day young embryo. The embryonic ectoderm (EE) and the extra-embryonic ectoderm (EXE) have differentiated and the proamniotic cavity is more prominent. 330×.

FIG. 12. Five-and-a-half-day senescent embryo. The proamniotic cavity is present and the proximal endoderm (PE) has completed its development, appearing similar to the young embryo after 5 days. 370×.

was also found in subsequent embryonic development. The proamniotic cavity appeared after 5 days in the young embryo (Fig. 9), while it did not appear until 5½ days in the senescent embryo (Fig. 12).

Of the total blastocysts studied only five appeared abnormal. A 3-day and a 4-day young blastocyst were degenerating, and three early senescent embryos, one at 4½ and two at 5 days, were morphologically different from their adjacent implants. Other blastocysts may have been abnormal but were not detectable if in a late stage of degeneration. The small number of blastocysts found prior to the Pontamine Blue reaction in young and senescent animals was undoubtedly due to the method of fixation and embedding. The entire uterus was sectioned in these animals.

Although an obvious lag of 12 h occurred during the implantation and embryonic development of senescent blastocysts, the sequence of events and morphology of these processes appeared similar to those of the young animal at the light microscope level.

DISCUSSION

The greatest loss of fertilized ova in senescent golden hamsters is reported to occur between 56 and 132 h of development (Connors *et al.*, 1972). Blastocysts from senescent animals, beginning at 3 days and developing through 3½, 4, 4½, 5, and 5½ days of pregnancy, exhibit a 12-h lag in implantation and embryonic development when compared with blastocysts from young hamsters. This lag does not result from delayed ovulation, since ovulation time is the same in young and senescent hamsters (Stockton *et al.*, unpublished). Even though young males were used in mating both groups, it is not known if changes occur within ova from senescent animals, making them inherently slower in development; or if the penetration of spermatozoa is impeded, causing delayed fertilization. In the latter case, hamster ova *in vivo* have been shown to become progressively defective cytogenetically if fertilization is delayed 3, 6, and 9 h following ovulation (Yamamoto and Ingalls, 1972). Female hamsters rarely mated 9 h after ovulation; and when they did, the ova were all defective. If ova from senescent females are not fertilized for several hours after ovulation, many of the embryos may be lost through developmental abnormalities.

The faint and delayed response of the senescent uterus to Pontamine Blue may imply an impaired circulation causing early embryonic death. Senger *et al.* (1967) found that cauterization of the uterine artery and vein in pregnant mice 1 and 5 days post-coitum had a significant effect on implantation. In contrast, Bruce (1972) has recently shown that ligating the uterine artery during early pregnancy (1, 2, or 7 days of gestation) had no apparent effect on litter size or fetal viability in rats. Further studies should be conducted to clarify this contradiction. Intrauterine oxygen tension is known to rise significantly immediately prior to implantation in rats (Yochim and Mitchell, 1968; Yochim, 1971), probably influencing hormonal effects on endometrial metabolism. If sufficient oxygen were not present because of poor capillary permeability, it could also be a limiting factor in implantation.

There is evidence that the uterus in senescent animals is not as responsive as that in young animals. Blaha (1964) found the number of decidual cells was less extensive in the uterus of the senescent hamster, compared with that of younger animals at the same stage of pregnancy. Blaha (1967), using several methods of artificially inducing a decidual response in older hamsters, also demonstrated a reduction in the capacity of older animals to develop deciduomata, as well as a reduction in the size of the deciduomata when formed. Finn (1966), using similar techniques on ovariectomized aged mice, supplied with exogenous hormones, found the same thing.

In the present study, formation of the decidual cells in older uteri was delayed in accordance with the delay in implantation. The extent of the reaction under the natural stimulus of the blastocyst appeared comparable to that in younger uteri 12 h earlier. Only three senescent embryos (5 and 5½ days) exhibited a decrease in the density of peripheral decidual cells mesometrially. One of the proposed functions of the decidual cells is to provide a source of nourishment for newly implanting embryos. It has also been suggested that uterine blood vessels may play an important role in transporting decidual nutrients to the early embryo (Finn, 1971). This evidence is based on electron micrographs, which show basement membranes of blood vessels extending from the decidua, as absent or incomplete, once they surround a mouse blastocyst on day 6 of pregnancy.

During implantation in rodents, the importance of synchrony between the blastocyst and the endometrium is well established (McLaren and Michie, 1956; Noyes and Dickmann, 1960; Doyle *et al.*, 1963; Noyes *et al.*, 1963). Although the senescent animal exhibits an asynchrony during implantation when compared with the young hamster, the sequence of events during implantation appears synchronous in both animals. Asynchrony may be most important in terms of metabolism. Weitlauf (1971) has recently demonstrated by radioautography the continuous incorporation of [^{35}S] methionine by hamster embryos during the preimplantation period. In ovariectomized animals, protein synthesis continued until the first signs of ovum degeneration on day 4. The hamster, unlike the mouse, does not undergo delayed implantation (Prasid *et al.*, 1960; Orsini and Meyer, 1962). If protein synthesis begins at the same time in young and senescent hamster blastocysts but the stages of cleavage and development are retarded in the older ova, some may exhaust their nutritive supply during early implantation.

If ova from senescent hamsters are fertilized at a similar time as those from young animals but are developmentally delayed from implanting for 12 h, their stored nutritive sources may be diminished. Once implantation has occurred, if an impeded circulation fails to provide sufficient oxygen and nourishment, the newly implanted embryo may be incapable of surviving. Histochemical and electron microscope studies are currently underway on these same stages of development to provide additional answers about reproductive failure in aging golden hamsters.

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